Smart Irrigation

Digital Twin for Precision Agriculture - Setup and Troubleshooting Guide

This prototype reads temperature, humidity, soil moisture, rain and light, calculates a Water Requirement Index (WRI), shows everything on a Blynk dashboard and on a local web page, and lets you switch a water pump on and off remotely. The controller is a **NodeMCU ESP8266**.

1. What you need

Part	Purpose
NodeMCU ESP8266 + USB cable	Main controller with Wi-Fi (powered over USB)
DHT11 module	Temperature and humidity
Soil moisture sensor + control board	Soil moisture (analog)
Rain sensor plate + control board	Rain / no rain (digital)
LDR + 10 k resistor (or LDR module)	Dark / bright (digital)
1-channel 5V relay module	Switches the pump's power
DC water pump + separate supply	Water delivery (5V/12V adapter, or 9V battery for short tests)
1N4007 diode, jumper wires, breadboard	Protects the board from motor spikes; connections

2. Sensor wiring (all sensors use 3V3, never 5V)

Component	Component pin	NodeMCU pin	Notes
DHT11 module	VCC (+) DATA (S / OUT) GND (-)	3V3 D5 GND	Check the printed labels; pin order differs between boards.
Soil moisture	VCC AO GND DO	3V3 A0 GND not used	The 2-wire probe connects to the control board in either direction.
Rain sensor	VCC DO GND AO	3V3 D6 GND not used	The plate's 2 wires go to the board in either direction. Tune sensitivity with the blue screw.
LDR (bare)	Leg 1 Leg 2	3V3 D7, and through a 10 k resistor to GND	Acts as a dark / bright switch. For an LDR *module*: VCC to 3V3, GND to GND, DO to D7, and set DARK_LEVEL to HIGH in the code.

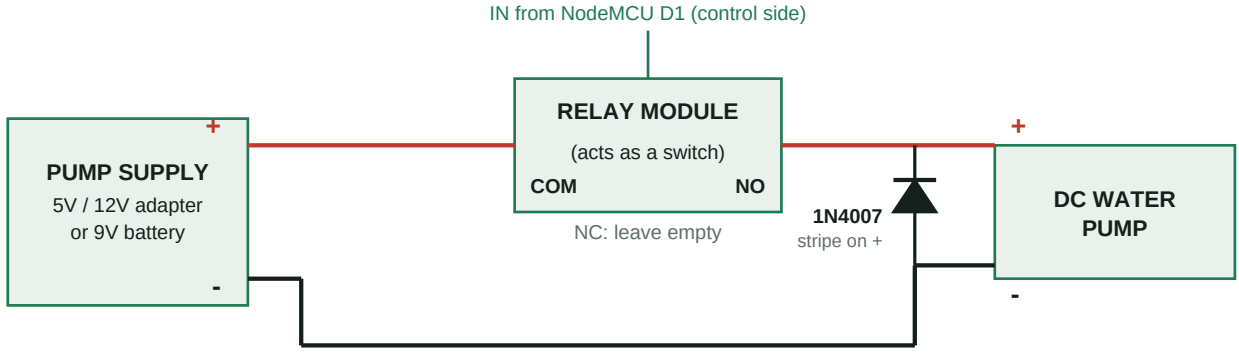
Pin summary: **D5** DHT11 | **A0** soil | **D6** rain | **D7** light | **D1** relay.

3. Relay and pump wiring

Control side (3 pins on the relay board, low voltage from the NodeMCU):

Relay pin	Connects to
VCC	NodeMCU **Vin** (5V from USB; the NodeMCU must be powered over USB)
GND	NodeMCU **GND**
IN	NodeMCU **D1**

Pump side (3-screw terminal, completely separate from the NodeMCU):



Connection	Wire
1	Pump supply **+** (battery red) to relay **COM**
2	Relay **NO** to pump **+** (red wire)
3	Pump **-** (black wire) to pump supply **-** (battery black)
4	**1N4007 diode** across the two pump wires, stripe (cathode) on the pump + side

Wiring the supply + to NO and COM to the pump + also works, because COM and NO are just the two ends of one switch. Always use NO and leave NC empty, so the pump stays off when the board is off or resetting.

> **Safety:** Low-voltage DC only (5V to 12V). Never connect mains voltage to this circuit. Never power the pump from the NodeMCU or its USB port. Do not join the pump supply's negative to the NodeMCU's GND; the relay keeps the two sides separate. Wire the pump with the supply unplugged, and never run the pump dry. Check the pump's rated voltage: a 3V to 6V pump should not be run on a 9V battery for more than a few seconds.

4. First power-up test (do in this order)

- Upload the sketch with the **pump circuit disconnected**. Open Serial Monitor (115200 baud) and check every sensor reading.
- Use the Blynk switch or the local web page and listen for the relay click. Click on = pump on, click off = pump off.
- Connect the pump circuit with the supply unplugged. Check wires and diode direction, then plug in.
- Put the pump in water and switch it on and off from the dashboard.

5. Blynk setup (one-time)

Use Blynk Cloud at **blynk.cloud**. Menu names can change slightly between versions.

1. **Create the template:** Developer Zone > My Templates > New Template. Name *Smart Irrigation*, hardware **ESP8266**, connection **WiFi**.
2. **Add datastreams** (Datastreams tab > New Datastream > Virtual Pin). The Virtual Pin number must match this table:

Name	Virtual pin	Type	Range	Unit
Temperature	V0	Double	0 - 60	deg C
Soil Moisture	V1	Integer	0 - 100	%
Rain	V2	Integer	0 - 1	-
Light Intensity	V3	Integer	0 - 100	%
Water Requirement	V4	Integer	0 - 100	%
Pump	V5	Integer	0 - 1, default 0	-
Humidity	V6	Integer	0 - 100	%
Soil Raw	V7	Integer	0 - 1024	-

Free plans may limit datastreams. If you hit the limit, leave out V7 first, then V6. The code still works.

3. **Build the dashboard:** open the template's Dashboard tab. Add Gauges for Temperature, Soil Moisture, Light Intensity, Water Requirement and Humidity; Labels for Rain and Soil Raw; and a **Switch** linked to Pump (ON = 1, OFF = 0). Use a Switch, not a push-Button. Click Save.
4. **Create the device:** Devices > New Device > From template > Smart Irrigation > Create. The device page shows **BLYNK_TEMPLATE_ID**, **BLYNK_TEMPLATE_NAME** and **BLYNK_AUTH_TOKEN**.
5. **Paste them** over the three lines at the very top of the sketch (they must stay above all #include lines), keeping the quotation marks.

6. Uploading the code

- **Arduino IDE:** File > Preferences > add `http://arduino.esp8266.com/stable/package_esp8266com_index.json` to Additional Boards Manager URLs. Then Boards Manager > install **esp8266 by ESP8266 Community**.
- **Libraries** (Library Manager): **Blynk** by Volodymyr Shymanskyy, **DHT sensor library** by Adafruit, **Adafruit Unified Sensor**. Restart the IDE after installing.
- **Board:** NodeMCU 1.0 (ESP-12E Module). Select the COM port, click Upload. If it will not connect, hold the FLASH button while it starts.
- Open Serial Monitor at **115200**. You should see sensor readings every 2 seconds and a line `Local control page: http://192.168.x.x`.

7. Local control page

On any phone or laptop connected to the **same Wi-Fi**, open the IP address from Serial Monitor (or `http://irrigation.local`). It shows live readings and a pump toggle, and works even if the internet is down. The page has no password, so use it only on a trusted network. The Poppins font needs internet to load; without it the page still works in a default font.

8. Changing settings in the code

All settings are near the top of the sketch. After any change, click Upload again.

Setting	What it does / how to set it
WIFI_SSID, WIFI_PASS	Your Wi-Fi name and password, in quotes. Must be a 2.4 GHz network.
BLYNK_TEMPLATE_ID, _NAME, BLYNK_AUTH_TOKEN	Copied from the Blynk device page (section 5).
SOIL_DRY_RAW, SOIL_WET_RAW	Calibration. Note the 'Soil raw' value in Serial Monitor with the probe in dry air (dry) and in a glass of water (wet), then enter them. Soil % depends on these.
RAIN_WET_LEVEL	LOW or HIGH. If it shows rain when dry, flip it.
DARK_LEVEL	LOW for a bare LDR divider, HIGH for most LDR modules. Flip if light is backwards.
TEMP_MIN, TEMP_MAX	Temperature that counts as 0% and 100% in the WRI. Defaults: 19 and 35 deg C.
RELAY_ACTIVE_LOW	true for most blue relay modules. Set to false if the relay is on when it should be off.
PUMP_MAX_RUN_MS	Automatic pump cut-off. Default 2 minutes. Example: (5UL * 60UL * 1000UL) for 5 minutes.
WIFI_GRACE_MS	Pump stops if Wi-Fi stays lost this long (default 10 seconds).
READ_INTERVAL_MS	How often sensors are read (default 2000 ms; the DHT11 needs at least 1000).
WRI formula (readSensors)	0.60 x soil dryness + 0.20 x temperature factor + 0.10 x light factor + 0.10 x rain factor. Change the weights there; keep them adding up to 1.0.

9. Troubleshooting

Problem	Likely cause and fix
Upload fails / no COM port	Use a data-capable USB cable (not charge-only), install the CH340 or CP2102 driver, try another port, hold FLASH during upload.
Error: DHT.h or BlynkSimpleEsp8266.h not found	Library not installed. Install it in Library Manager and restart the IDE. Board must be NodeMCU 1.0.
Serial Monitor shows only dots at Wi-Fi	Wrong name or password (case sensitive), or a 5 GHz network. Use 2.4 GHz.
Wi-Fi works, Blynk stays offline	Re-copy the three BLYNK_ lines from the device page. They must be above the #include lines. Check that the network has internet.
Blynk online but widgets are empty	Widgets not added or not linked to a datastream. Also, values are sent only when they change or once a minute, so wait up to 60 seconds.
DHT: READ FAILED	Check the module's pin order (VCC, DATA, GND), DATA on D5, and that the DHT library is installed.
Soil raw always 1024 or 0%	AO not connected to A0, or sensor not powered from 3V3. Check the control board's wires. Then calibrate.
Rain or light reading is backwards	Flip RAIN_WET_LEVEL or DARK_LEVEL (section 8).
Relay does not click	Check Vin to VCC, GND to GND, IN to D1. The NodeMCU must be powered over USB (Vin gives 5V only then).
Relay clicks the wrong way round	Set RELAY_ACTIVE_LOW to the opposite value.
Relay clicks but pump does not run	Check the pump-side wiring (COM, NO), that the supply is plugged in, battery charge, and that the pump's voltage matches the supply.
Pump stops after 1 to 2 seconds, or board resets	Add the 1N4007 diode across the pump (stripe on +). Use a good USB charger and cable. Replace a weak 9V battery. Serial Monitor shows the reason (Wi-Fi lost, or a restart).
Pump stops by itself after 2 minutes	That is the safety cut-off. Change PUMP_MAX_RUN_MS if you need longer.
Switch turns off as soon as released	Blynk widget is a Button in push mode. Use a Switch widget.
Local page will not open	Phone must be on the same Wi-Fi. Use the IP from Serial Monitor; irrigation.local does not work on every device.

> **Note:** The WRI is an estimated index, not an exact water volume. Calibrate the soil values and temperature range against real crop and soil conditions before relying on it. Keep your Wi-Fi password and Blynk token out of any code you share. If a token leaks, reset it in the device's settings in Blynk.